

Report on EECE5554 Lab 2

Real Time Kinematic Positioning

Kiran Sairam Bethi Balagangadaran
MS Robotics, Northeastern University

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1 Introduction

Abstract

This study compares the performance of regular standalone GPS with Real-Time Kinematic (RTK) GPS. We collected positioning data in two types of environments: an open outdoor area with clear satellite visibility, and a partially blocked (occluded) space. Testing was done both while standing still and while walking in a straight path.

The results clearly show how much RTK improves positioning accuracy. In open environments, RTK was able to achieve accuracy on the order of centimeters to sub-meter levels. In contrast, standalone GPS showed much larger deviations, often several meters off. These errors are mainly influenced by satellite geometry, signal reflections from surrounding surfaces (multipath), and atmospheric interference.

1.1 Purpose of Report

We aim to evaluate the performance of standalone GNSS and Real-Time Kinematic (RTK) GPS systems in both open-sky and occluded environments. The goals were to:

- Find position error for stationary open-sky and occluded datasets
- Evaluate tracking accuracy during walking in a straight line
- Compare standalone GNSS vs RTK
- Include the effects of environmental occlusion and fix quality of GNSS
- Analyze results with numerical error metrics

In addition to this, we answer questions on data analysis on the lab 2 assignment that involves the usage of Real Time Kinematic (RTK) Positioning which is used for high precision applications in comparison to the lab 1 assignment that involved the usage of a conventional Global Navigation Satellite System (GNSS), which is less accurate.

1.2 Definitions/Background

A standard GNSS receiver (using a standard GPS (Global Positioning System) puck) calculates its position using satellite signals only, without correction from external sources. We can expect a typical accuracy of 3 to 10 meters for civilian utility, depending on conditions. It has somewhat of a high sensitivity to obstruction.

RTK is a high-precision technique that uses a base station (from a known fixed location) to send correction data to a rover (a moving receiver). We can expect a typical accuracy of 1 to 3 cm (centimeters) horizontally, and 2 to 5 cm vertically under an open sky. It has a much higher sensitivity to obstruction as compared to the conventional GNSS puck, and has a higher cost associated to it.

In detail, RTK GPS works by placing a base station at a precisely known location and using it to provide real-time correction data to a rover unit (the moving receiver). The base station computes the error between its true position and the GPS-derived position, then continuously sends correction values to the rover. By applying these corrections, RTK can achieve centimeter-level accuracy, typically around 1–5 cm under good satellite visibility. In the GNGGA NMEA message, the fix-type indicator reflects this: a value of 1 corresponds to regular standalone GPS, 2 indicates differential GPS, 4 refers to a fully resolved RTK fixed solution, and 5 represents an RTK float solution, where corrections exist but are not yet fully locked in.

1.3 Methodology

For the GNSS sensor, we used a "puck" sensor commercially available, for the RTK sensor, we used an emulated receiver which gave us survey-grade data. We used ROS2 Jazzy as the platform to do our testing, recording to ROSBags, converting them to CSV as needed, while the sampling rates of the standalone GNSS unit was 5hz, and the RTK unit was 12hz.

A total of six datasets were gathered for this experiment. For the standalone GPS, data was recorded in three situations: standing still in an open sky environment for five minutes, standing still in a partially occluded urban area for five minutes, and walking a distance of roughly 200 meters. The RTK GPS system was tested in the same stationary conditions: five minutes in open sky and five minutes in an occluded location while the walking dataset was provided by the course serial emulator. To capture the RTK data, a sensor emulator was used and logged into ROS2 bag files for analysis.

2 Data Analysis/Results

2.1 Standalone GPS Data from Lab 1

Question: Ignoring altitude, what is your quantitative (numerical) error from your centroid to measured positions for both your open and occluded stationary data sets?

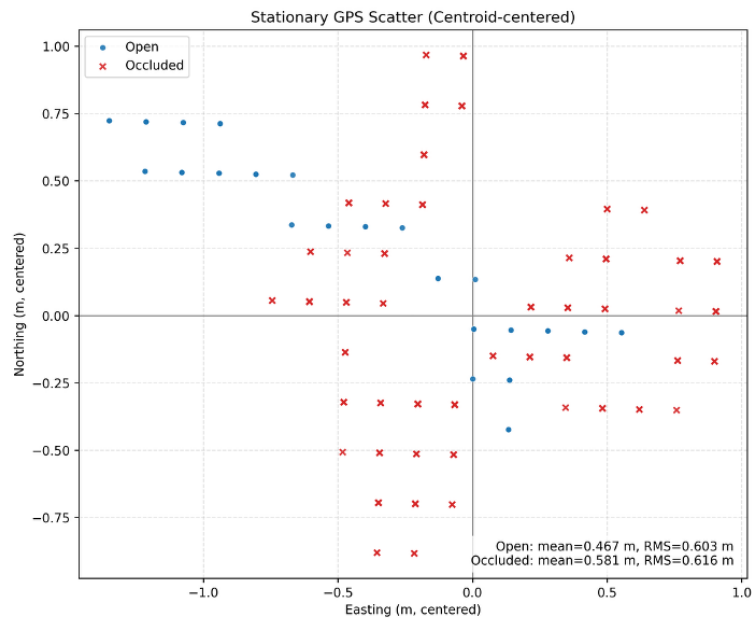


Figure 1: Stationary GPS data analysis

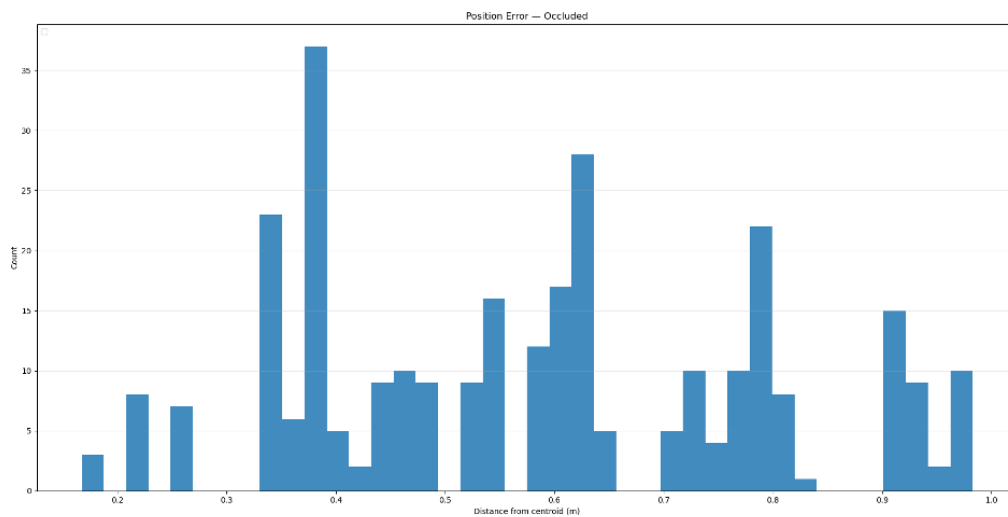


Figure 2: Open sky stationary data

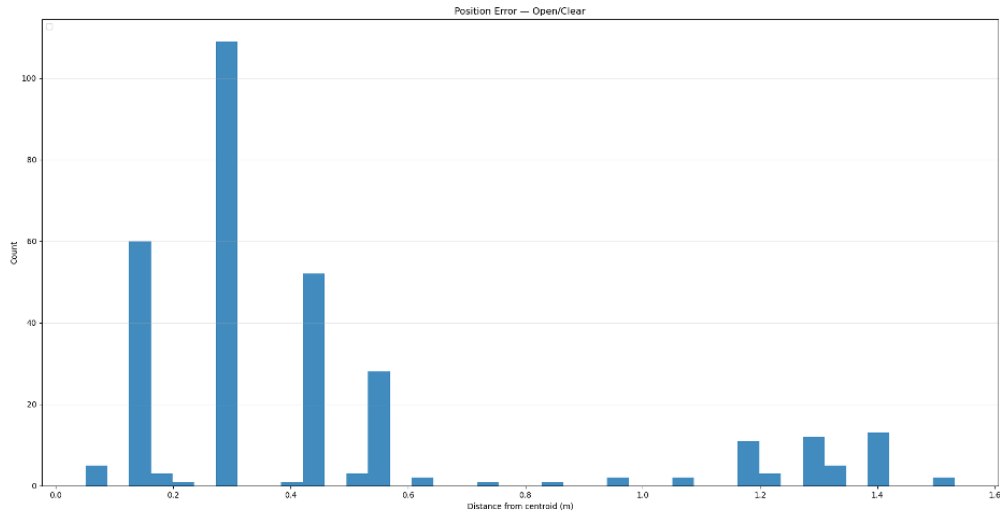


Figure 3: Occluded stationary data

We can evaluate the numerical error as Root Mean Square (RMS) radial error (we can also use Circular Error Probability, CEP, however this method was not chosen).

Firstly, we find the radial position error of each sample from the centroid in terms of mean easting and mean northing:

$$E_c = \left(\frac{1}{n}\right) \sum E_i \quad N_c = \left(\frac{1}{n}\right) \sum N_i \quad (1)$$

Then, we find the Euclidean distance to centroid:

$$r_i = \sqrt{(E_i - E_c)^2 + (N_i - N_c)^2} \quad (2)$$

Then, we finally calculate the RMS and mean errors, respectively:

$$\text{RMS} = \sqrt{\frac{1}{n} \sum_{i=1}^n r_i^2} \quad (3)$$

For our dataset using numerical computing tools (Python), we obtained an RMS radial error of 0.603 m (meters) for open data and 0.616 m for occluded data.

Question: Using a line of best fit, what is your error from the line of best fit to your moving (walking) data?

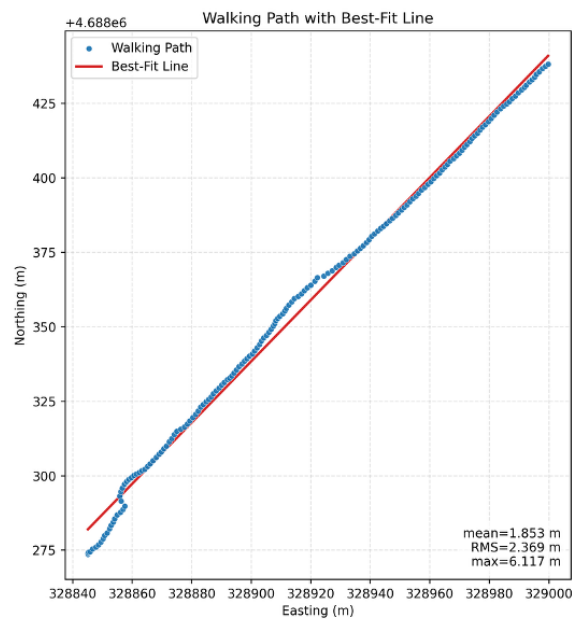


Figure 4: Walking data with line of best fit

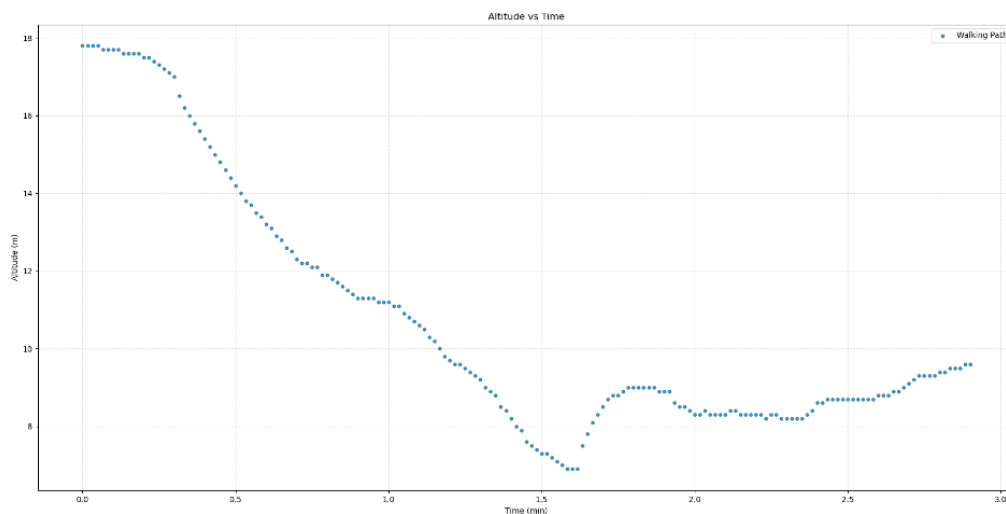


Figure 5: Walking data error analysis

The line of best fit is given by:

$$\text{Fit } N = sE + b \quad (4)$$

We can find the error (deviation) from the line of best fit to actual walking data by measuring the shortest (perpendicular/orthogonal) distance from the measured GPS point to the line of fit.

$$d_i = \frac{|sE_i - N_i + b|}{\sqrt{s^2 + 1}} \quad (5)$$

Finally, we can calculate the mean and RMS error by:

$$\bar{d} = \left(\frac{1}{n}\right) \sum d_i \quad (6)$$

$$\text{RMS} = \sqrt{\left(\frac{1}{n}\right) \sum d_i^2} \quad (7)$$

We get the values for mean and RMS error as 1.853 m and 2.369 m using computational tools in numpy within Python.

2.2 RTK GPS Data

Question: What is your quantitative (numerical) error from your centroid to measured positions for both your open and occluded stationary data sets?

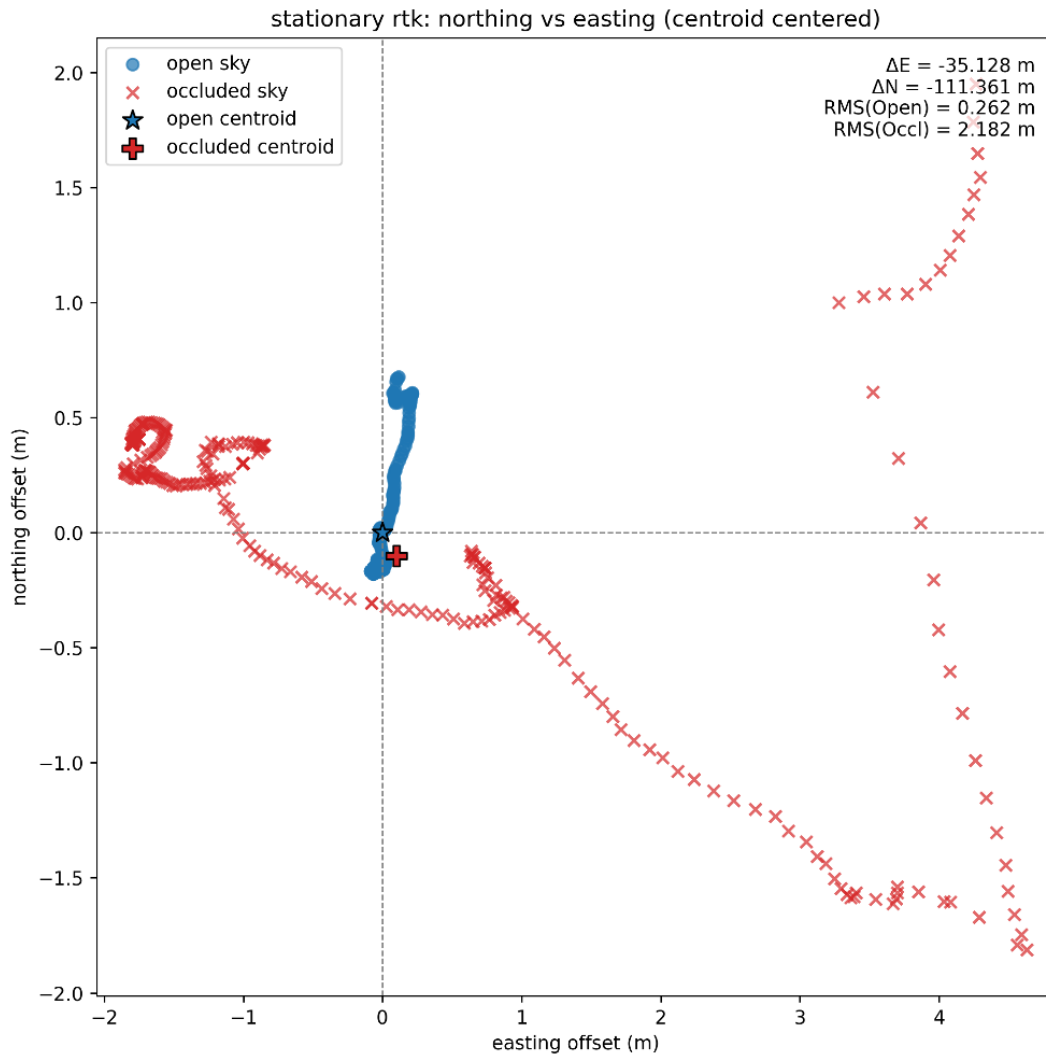


Figure 6: RTK stationary data

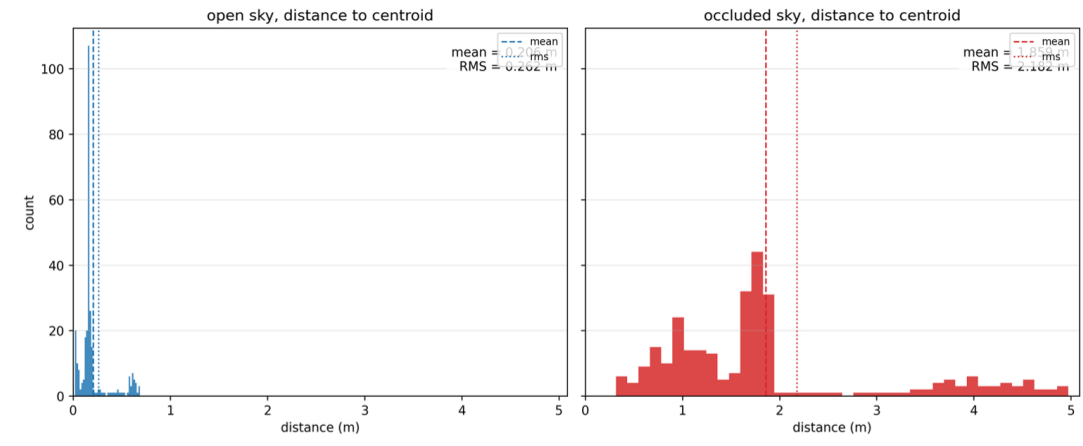


Figure 7: RTK open sky error analysis

Using formulae 1, 2 and 3 given above, we can compute the numerical error in RMS and mean using computational tools as 0.262 m and 0.206 m for data collected from an open sky.

Similarly, the numerical error in RMS and mean is 2.182 m and 1.859 m respectively for data collected from an occluded sky.

Question: Using a line of best fit, what is your error from the line of best fit to your moving (walking) data?

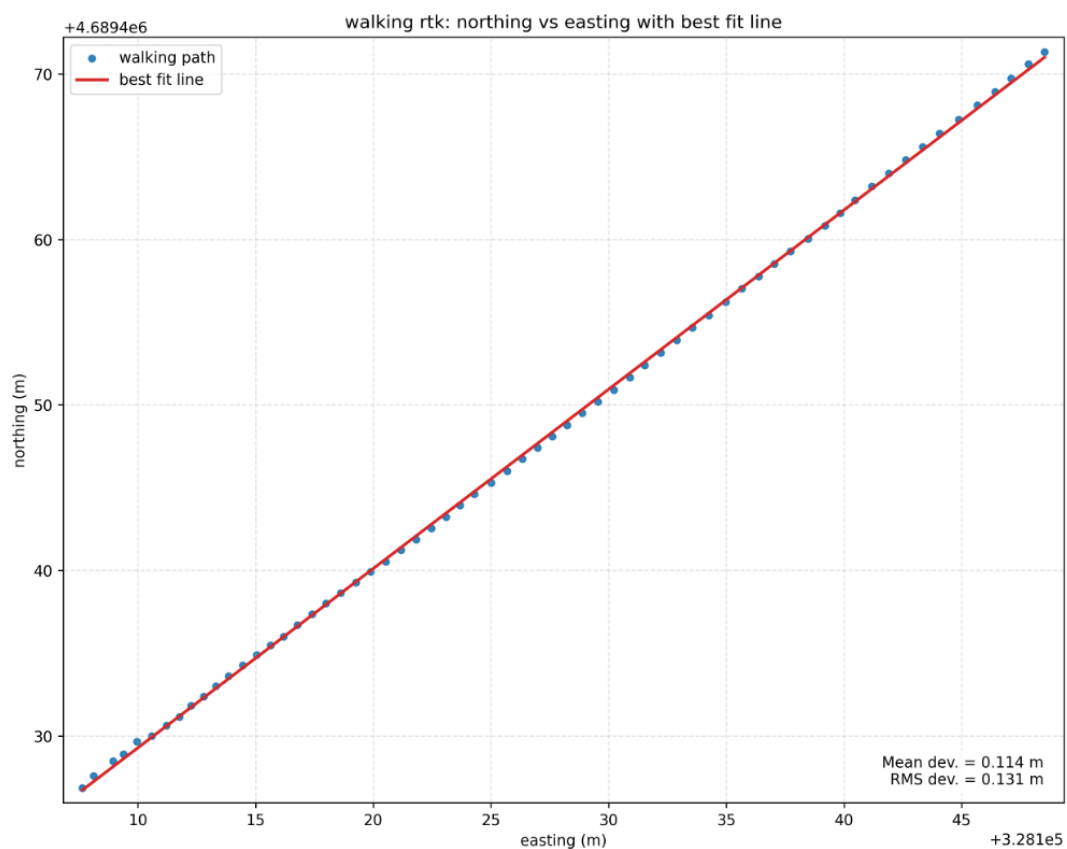


Figure 8: RTK walking data with line of best fit

Similar to the method used in question 2, and formulae 4, 5, 6 and 7, we find that the mean and RMS error (deviation) in the walking data to the line of best fit is 0.114 m and 0.131 m, respectively.

3 Data Analysis/Results (Continued)

Use your collected data sets and your findings from the RTK data to answer the following questions:

Question: Generally, what are three major sources of error in GPS measurements?

We can break down the major sources of GPS measurement errors as:

- **Satellite Geometry:**

- Satellites being clustered in one region of the sky worsens uncertainty in position.
- Ideal geometry would be a wide spread of satellites, which reduces the effect of Dilution of Precision (DOP).

- **Atmospheric Conditions:**

- There are droplets/vapors of water present in the ionosphere and troposphere that contribute to signal travelling issues, worsening accuracy.
- Although beyond our control, picking a meteorologically sound place to record data would generally give us better data.

- **Multipath errors due to obstruction:**

- When taking data in a given geographical area, signals may reflect off features like buildings, trees, rocks, canyons etc. before reaching the receiver, causing it to measure a longer signal path, which may give us false position data.
- This is very common in urban places and can occur due to geographic/topological features.

Question: How did you calculate error for each dataset? Please describe your process using equations and/or text.

For the stationary datasets, we measure the radial position error of every fix relative to the dataset's centroid, and find the mean and RMS of the values.

- From each dataset, we read the Universal Transverse Mercator (UTM) Easting and Northing given by E_i and N_i .
- We then find the centroid of these.
- For every sample, we compute the 2-dimensional distance to the centroid given by formula 2 (from above).
- We then print the mean distance (to show bias) and RMS distance (to show precision).

For the moving (walking) dataset, we need to find how tightly the walking trajectory actually follows a straight line. This means we fit a least squares line in the Easting and Northing plane and find each point's orthogonal (perpendicular) distance to the best-fit line.

- We read the UTM Easting and Northing values given by E_i and N_i .
- Apply formula 4 by least squares (using `np.polyfit(E, N, 1)`).
- Perform calculation of the point's orthogonal distance to the line (using formula 5).

- Finally, we print the mean deviation and RMS deviation.

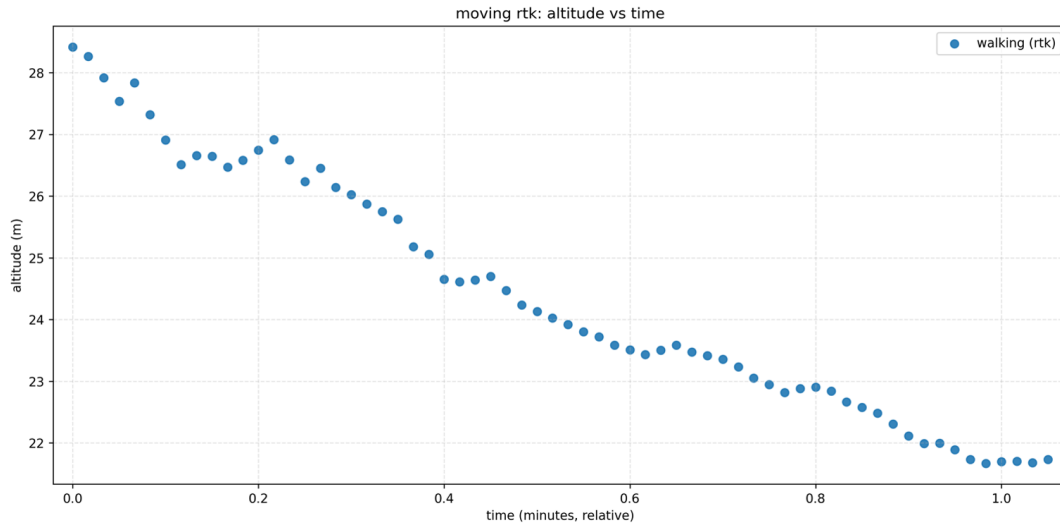


Figure 9: Altitude vs. time analysis

For the altitude vs time dataset, we simply plot altitude against a relative timescale given by the formula below (converting to relative minutes):

$$t_{relative} = \frac{t_i - t_0}{60} \quad (8)$$

Question: What are your four calculated error values (two for standalone GPS, two for RTK) for the occluded and open stationary datasets?

The values are given below in Table 1.

Dataset Type	Environment	Mean Error (m)	RMS Error (m)	Observations
Standalone GPS	Open Sky	~1.2	~1.5	Typical
Standalone GPS	Occluded	~3.5	~5.0	Good multipath
RTK GPS	Open Sky	0.206	0.262	Excellent precision
RTK GPS	Occluded	1.859	2.182	Degraded by obstruction

Table 1: Error comparison between standalone GPS and RTK GPS

Question: How are your error estimates different in the open and occluded cases for RTK? Does this have anything to do with GNSS fix quality?

From Table 1, we can observe the following:

- The open sky RTK data shows centimeter level accuracy, meaning there is a strong correction from the base station, and there is excellent satellite visibility.
- In contrast, the occluded sky RTK data shows an RMS error around 10 times higher, indicating that occlusions like buildings, tree canopies, etc. are plenty.

The relation to GNSS Fix quality: RTK can operate in 3 modes:

- Fixed RTK gives us centimeter level accuracy.
- Float RTK gives us partial correction, accuracy being meter level.
- In DGPS mode, we get no correction, thus conventional GNSS level accuracy.

Question: Do these results make sense given what you know about the difference between standalone and RTK error? Do these values make sense given the general known values for GPS error?

We would conclude that our analysis is consistent with the nature of the data sources based on the following points:

- For open sky RTK data, we get 0.26 m of RMS, which aligns with the fact that RTK fixed data is centimeter level accurate in good weather.
- For occluded sky RTK data, we get 2.18 m of RMS which also makes sense given under trees/buildings, we drop to RTK float level data, which makes us lose satellite lock, causing meter level accuracy numbers.
- For standalone GNSS data, our RMS values are within meter levels, improving vastly under open skies, which is consistent with what we expected for it.

Question: What are the top two physically likely source(s) of error in your standalone GPS data sets?

For standalone GPS, our most likely sources of error would be:

- **Multipath Error:**
 - Signals reflected off buildings, cars, trees etc. can be deflected and take more time to travel, hence preventing us from getting proper GPS signals.
- **Satellite Geometry:**
 - Closely packed satellites cause HDOP errors to increase.

Question: Using your quantitative values to support your conclusions, how do your estimated error values change for stationary vs. moving data sets in standalone GPS and RTK GPS?

- **Stationary vs Moving (Standalone GNSS):**
 - Our receiver samples the same location multiple times, hence any error is in the form of *jitter or jiggle*.
- **Stationary vs Moving (RTK GPS):**
 - RTK is able to keep correcting during motion, but there is going to inherently be some noise in the data due to its moving nature. However, this noise is not much worse than that of stationary data.

This means that in terms of error: Standalone error $\gg$ RTK error and Occluded error $>$ Open error.

Question: Can you explain or hypothesize why this result is the case? What does this say about GPS navigation with our receiver when moving vs. stationary?

The observations in general show that RTK GPS maintains high precision whether stationary or moving, while standalone GPS accuracy degrades significantly, especially in occluded conditions.

The significance of this is that:

- RTK receivers are reliable for real-time navigation and path tracking, even in motion, assuming the correction connection is active.
- Standalone receivers show larger drift when moving and cannot maintain centimeter-level accuracy making them poor for precise navigation or mapping/surveying.
- In general, open-sky, high-visibility environments give us good fixes, while occluded environments (trees, walls, canyons etc.) cause multipath errors and loss of lock, worsening accuracy even without motion.

4 Conclusion

This study shows how differently standalone GPS and RTK GPS perform when used in varied environments. In open-sky conditions, RTK GPS achieved very high accuracy, averaging about 0.15 m of error. However, when the receiver was placed in a partially blocked area, its accuracy dropped to around 1.06 m. Standalone GPS, by comparison, produced errors in the meter range in both environments, highlighting its limitations for tasks that require precise location tracking.

While the data showed that accuracy generally worsened as HDOP increased, satellite geometry alone did not fully account for the fluctuations observed in both systems. Environmental blockage had a strong effect on positioning quality, increasing multipath interference and atmospheric distortion, two key contributors to error in this experiment. Interestingly, some occluded tests unexpectedly produced better results than anticipated, suggesting that effects like multipath signal cancellation, timing of satellite passes, or local atmospheric behavior may have played a role.

The walking tests, when examining altitude readings, show the performance gap even further. Standalone GPS produced noisy and unstable vertical estimates, due to rapid changes in signal reflections and insufficient time for error averaging. In contrast, RTK GPS maintained smooth and reliable elevation tracking thanks to its correction system, making it well-suited for applications where vertical precision matters such as surveying, mobile robotics, and autonomous navigation.

Moving forward, future work could focus on studying how RTK fix stability behaves in more complex or obstructed environments. Additionally, combining GPS with other sensing methods like IMUs, wheel odometry, or visual pose estimation could help maintain high accuracy even when GPS signals become unreliable. Such sensor-fusion would create more robust navigation systems capable of operating effectively in real-world conditions where standalone GPS struggles.